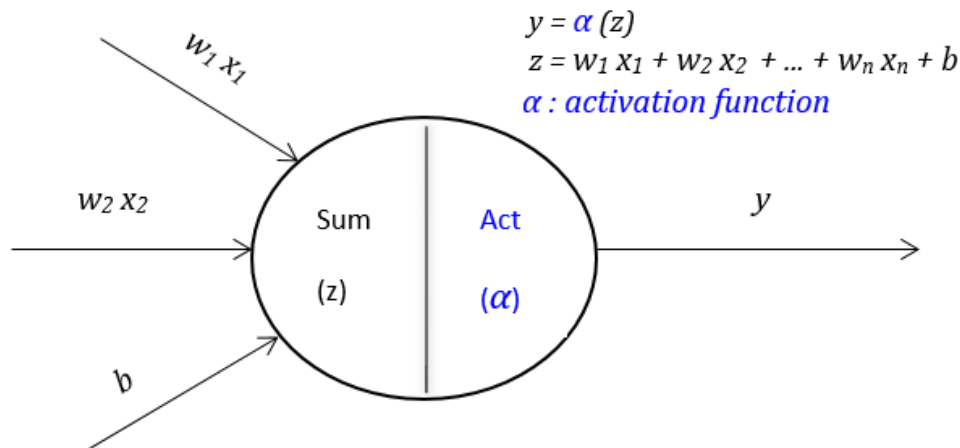


Activation Function

In artificial neural network each neuron forms a weighted sum of its inputs and passes the resulting scalar value through a function referred to as an activation function or transfer function.



Why Activation Functions Are Needed:

Non-linearity: Without them, the network behaves like a linear regression model, no matter how many layers it has.

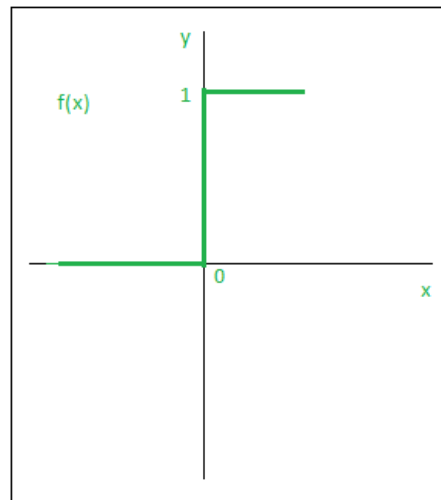
Learn complex patterns: They help the model learn and represent complex data like images, speech, etc.

Types of Activation Functions

Step Function: Step Function is one of the simplest kind of activation functions. In this, we consider a threshold value and if the value of net input say y is greater than the threshold then the neuron is activated. Mathematically,

$$f(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

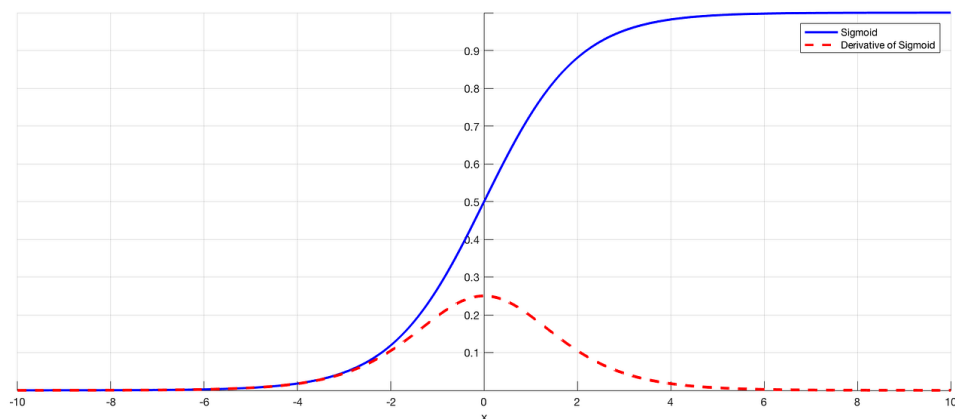
Graphical representation of Step function



Sigmoid function: The Sigmoid function is a type of activation function commonly used in neural networks. It is smooth and continuous, with an S-shaped curve, which makes it easy to differentiate — a crucial requirement for training models using backpropagation.

Mathematically, it is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Advantages

Non-linearity – Introduces non-linear behaviour, enabling the model to learn complex patterns.

Probability Interpretation – Outputs between 0 and 1, making it suitable for probability-based predictions.

Differentiable – Has a well-defined derivative everywhere, allowing gradient-based optimization.

Disadvantages

Vanishing Gradient Problem – For very large or very small inputs, gradients become near zero, slowing or stopping learning.

Not Zero-Centered – Output ranges from 0 to 1, causing gradients to have consistent sign and leading to inefficient weight updates.

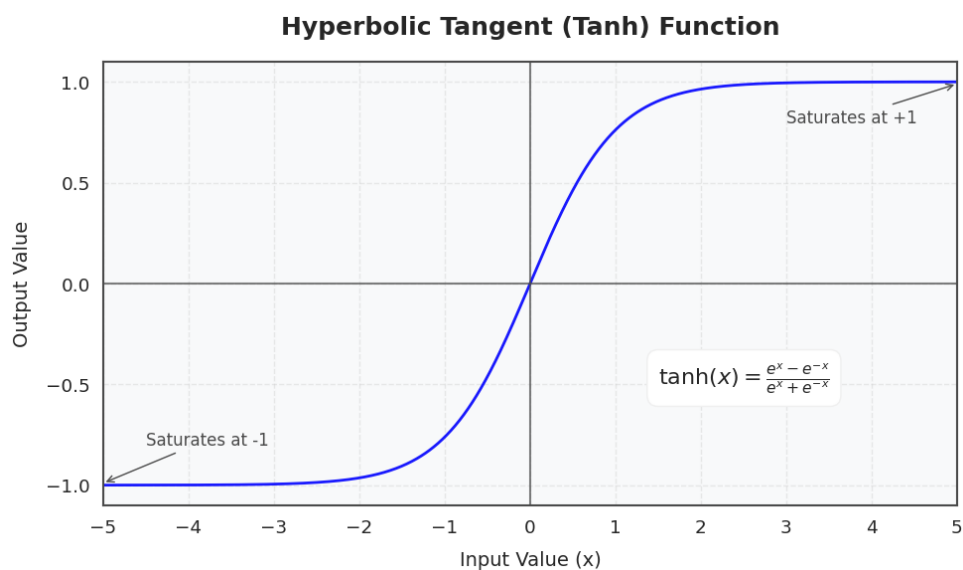
Computationally Expensive – Involves exponential calculations, making it slower compared to ReLU.

Vanishing Gradient Problem

The vanishing gradient problem occurs when gradients become very small during backpropagation in deep neural networks, especially with activation functions like sigmoid or tanh. This causes earlier layers to learn extremely slowly or stop learning, making the model hard to train.

Tanh

The **tanh (hyperbolic tangent) function** is an activation function that maps inputs to values between -1 and 1.



Advantages

Non-Linearity – Introduces non-linear behavior, enabling the model to learn complex patterns.

Differentiable – Has a well-defined derivative everywhere, allowing gradient-based optimization.

Zero-Centered – Outputs range between -1 and 1, which helps balance positive and negative values for faster training.

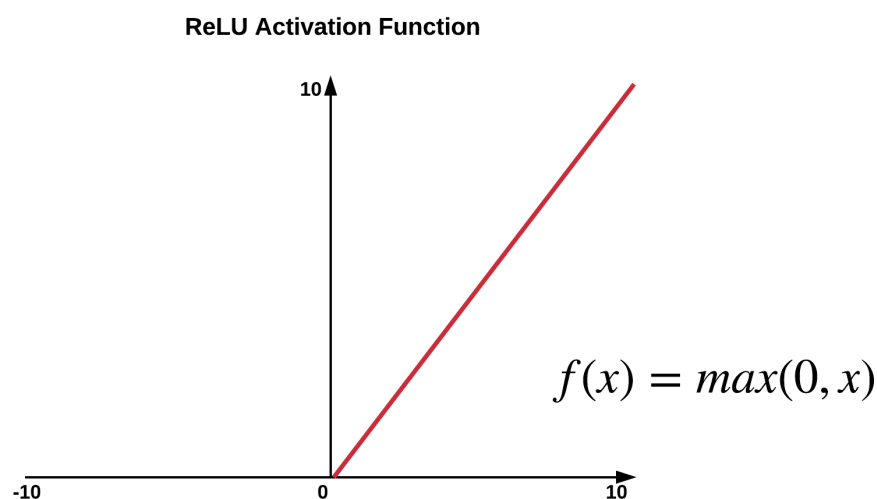
Disadvantages

Vanishing Gradient – For very large or small inputs, gradients approach zero, slowing down learning.

Computational Cost – Involves exponential operations, making it slower compared to ReLU.

ReLU

The **ReLU (Rectified Linear Unit)** is an activation function that outputs the input directly if it is positive, otherwise returns zero, making it simple and efficient for deep learning models.



Advantages

Non-Linearity – Adds non-linear behavior, enabling the model to learn complex relationships.

Computational Efficiency – Simple thresholding operation ($\max(0, x)$), making it faster to compute.

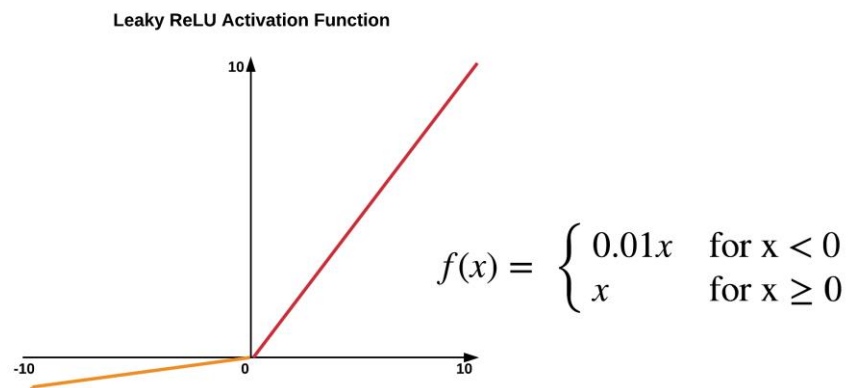
No Saturation for Positive Values – Avoids vanishing gradient for positive inputs, supporting deep network training.

Disadvantages

Dying ReLU Problem – Neurons can become inactive (output always 0) if inputs are negative, leading to no learning.

Leaky ReLU

The **Leaky ReLU** is an activation function that improves upon ReLU by allowing a small, non-zero slope for negative inputs, preventing neurons from dying and staying inactive.



Advantages

Non-Linearity – Allows the network to learn complex patterns like other non-linear activations.

Solves Dying ReLU Problem – Provides a small slope (e.g., $0.01x$) for negative inputs instead of zero, keeping neurons active.

Efficient Computation – Similar to ReLU, involves simple thresholding with a small adjustment for negatives.

Disadvantages

One disadvantage of **Leaky ReLU** is that the choice of the negative slope (commonly 0.01) is arbitrary and may not be optimal, making the model sensitive to this hyperparameter.

Optimizers

Optimizers are algorithms used to **update the weights and biases** of a neural network to minimize the loss function during training.

Types of Optimizers

Gradient Descent (GD)

Stochastic Gradient Descent (SGD)

Mini-Batch Gradient Descent

Momentum

Adam